

The Description of Additional Supplementary Files

Supplementary Video 1: Video showing ring-polymer molecular dynamics simulation of a Li^+ ion as it solvates, diffuses and then reacts with a Bz_2 molecule in the center of a 3600 He atom droplet. The molecules are initially tilted by 10° relative to the initial trajectory of Li^+ .

Supplementary Video 2: Video similar to Supplementary Video 1, but for an 1800 He droplet and Bz_2 initially aligned perpendicular to the trajectory of Li^+ . Note that the molecular dimer first rotates and then reacts to form $\text{Li}+\text{Bz}_2$, showing the importance of the stereodynamics of the reaction.